

SQL Fundamentals

Updated Exercises

Exercise 1 SQL Fundamentals : SELECT & FILTERING

1. Selecting & Sorting

01. Retrieve all columns from employees table

```
SELECT *  
FROM employees;
```



id	first_name	last_name	department	salary	hire_date	city
1	Alice	Green	IT	70000	2020-01-10	Johannesburg
2	Brian	Lee	HR	45000	2019-03-22	Cape Town
3	Cathy	Zulu	Finance	65000	2018-07-18	Durban
4	David	Mokoena	Marketing	50000	2021-11-05	Pretoria
5	Eva	Naidoo	IT	72000	2017-09-30	Johannesburg

02. Find all unique departments

```
SELECT DISTINCT department  
FROM employees;
```



department
IT
HR
Finance
Marketing

03. Retrieve first and last names ordered by salary descending

```
SELECT first_name,  
       last_name  
FROM employees  
ORDER BY salary DESC;
```



first_name	last_name
Eva	Naidoo
Alice	Green
Cathy	Zulu
David	Mokoena
Brian	Lee

04. Retrieve top 3 highest-paid employees

```
SELECT id,  
       first_name,  
       last_name,  
       salary  
FROM employees  
ORDER BY salary DESC  
LIMIT 3;
```



id	first_name	last_name	salary
5	Eva	Naidoo	72000
1	Alice	Green	70000
3	Cathy	Zulu	65000

2. Filtering with WHERE

05. Find employees in the IT department

```
SELECT id,  
       first_name,  
       last_name,  
       department  
FROM employees  
WHERE department = 'IT';
```



id	first_name	last_name	department
1	Alice	Green	IT
5	Eva	Naidoo	IT

06. Find employees in finance with salary > 60000

```
SELECT id,  
       first_name,  
       last_name,  
       department,  
       salary  
FROM employees  
WHERE department = 'Finance'  
AND salary > 60000;
```



id	first_name	last_name	department	salary
3	Cathy	Zulu	Finance	65000

07. Find employees in HR or Marketing

```
SELECT id,  
       first_name,  
       last_name,  
       department  
FROM employees  
WHERE department IN ('HR', 'Marketing');
```



id	first_name	last_name	department
2	Brian	Lee	HR
4	David	Mokoena	Marketing

NB The OR statement will give same results

08. Find employees not in IT

```
SELECT id,  
       first_name,  
       last_name,  
       department  
FROM employees  
WHERE department NOT IN ('IT');
```



id	first_name	last_name	department
2	Brian	Lee	HR
3	Cathy	Zulu	Finance
4	David	Mokoena	Marketing

09. Find employees in IT, HR or Finance using IN

```
SELECT id,  
       first_name,  
       last_name,  
       department  
FROM employees  
WHERE department IN ('IT', 'HR', 'Finance');
```



id	first_name	last_name	department
1	Alice	Green	IT
2	Brian	Lee	HR
3	Cathy	Zeile	Finance
5	Eva	Naidoo	IT

10. Find employees in IT with salary > 65000 and city Johannesburg

```
SELECT id,  
       first_name,  
       last_name,  
       department,  
       salary,  
       city  
FROM employees  
WHERE department = 'IT'  
AND salary > 65000  
AND city = 'Johannesburg';
```



id	first_name	last_name	department	salary	city
1	Alice	Green	IT	70000	Johannesburg
5	Eva	Naidoo	IT	72000	Johannesburg